The Rationality Horizon: Redefining Intelligence Beyond Human-Centric Metrics

Synthesizing interdisciplinary insights to propose a dynamic boundary for AI rationality

ABSTRACT

> "When machines cross our horizon, the real question is not what they can decide, but what we can still understand."

Artificial intelligence challenges inherited notions of rationality from economics, philosophy, and psychology. This paper synthesizes interdisciplinary insights to propose the **Rationality Horizon**—a dynamic boundary where AI rationality is measured across decision-theoretic optimality, cognitive realism, and contextual adaptability. Current benchmarks reflect the **GenAI Evaluation Paradox**, where training objectives diverge from evaluation criteria.

1. Philosophical Foundations of AI Rationality

AI systems achieve superhuman performance in narrow domains while showing limitations in generalization. Traditional rationality emphasizes expected utility maximization under perfect information. Reinforcement learning agents demonstrate perfect play in structured environments like Go.

Philosophical analysis identifies tensions in defining agency for machines. Outputs from AI systems represent probabilistic distributions rather than explicit decisions or beliefs.

2. Survey of (Ir)rationality in AI Research

> "Between noise and bias lies a third category: the behaviors our metrics never thought to look for."

AI rationality research spans definitions, benchmarks, human modeling, and challenges. Definitions vary: economics focuses on global optimality, psychology includes noisy heuristics, philosophy examines epistemic justification.

Benchmarks cover economic games such as ultimatum and prisoner's dilemma, alongside cognitive tests like the Wason selection task. Generative systems lack unified metrics, as outputs constitute distributions rather than discrete choices.

Human modeling shows AI performance exceeding humans on isolated tasks but underperforming in social contexts requiring theory of mind. Research distinguishes systematic biases from stochastic noise. **Bounded optimality** addresses rationality under resource constraints.

3. The GenAI Evaluation Paradox

> "Fluency without fidelity is the new uncanny valley of machine rationality."

Generative AI training minimizes prediction loss, producing fluent outputs that diverge from human evaluation standards. Planning tasks yield coherent strategies without iterative refinement to optimality.

Multi-agent systems reveal further divergence. Decentralized environments favor correlated equilibria over Nash equilibria in some cases. Human rationality supports cooperation; AI systems require equivalent capabilities.

4. LLM Self-Perception Analysis

> “Overconfidence in self-like opponents is not self-awareness; it is statistics wearing a mask of ego.”

Testing of 28 LLMs (GPT-4o, Claude 3.5 Sonnet, Llama-3.1 405B, etc.) across 4,200 trials in the **"Guess 2/3 of Average"** game reveals strategic opponent differentiation. Recursive reasoning converges to Nash equilibrium at zero.

Key results: against humans, models average ~20 (L1-L2 human reasoning levels); against "other AI models," ~0 (full rationality assumption); against "AI models like you," even lower (self-superiority positioning). 75% of advanced models exhibit this emergent hierarchy: humans < generic AIs < self-like AIs. This pattern appears in the referenced study.

Kim, K. H. (2025). Emergence of AI self-awareness measured through game theory. arXiv preprint arXiv:2511.00926.
The **AI Self-Awareness Index (AISAI)** quantifies this bias: models systematically underrate self-similar opponents while overcompensating against humans, suggesting training-induced overconfidence rather than true metacognition.

REFERENCES

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“Intelligence is no longer what humans are best at; it is what our benchmarks struggle to measure.”